

Controller shift in centered-world semantics

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1 Introduction

1.1 Overarching questions

- What would be a semantic account of control, which can capture the different controller choices, including the controller shift phenomenon?
- What do control facts tell us about the semantics of attitude reports?

Conclusions to be made

- Controller choice is constrained in two independent ways:
 - (i) *De se* requirement that the individual argument of the complement (i.e. PRO) must correspond to an attitude center.
 - (ii) The matrix controller argument corresponds to the embedded agent.
- The facts about controller choice give a new piece of evidence for the centered-world analysis of control.

1.2 Control shift

‘Ordinary’ control examples

- (1) a. John_i promised Mary PRO_i to leave. (Subject control)
b. John asked/persuaded/begged Mary_i PRO_i to leave. (Object control)

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Controller-shift (CS) examples (Hust and Brame 1976)

(2) (Underlying) Subject → Object control shift

- a. Mary_i was never promised [PRO_i to be allowed to leave].
- b. John never promised Mary_i [PRO_i to be allowed to leave].
- c. Grandpa promised the children_i [PRO_i to be able to stay up for the late show].
- d. Montana_i was promised (by the doctor) [PRO_i to be healthy by game time on Saturday]
(Examples are from Hust and Brame 1976, Farkas 1988, Sag and Pollard 1991, Landau 2000 here and below)

(3) (Underlying) Object → Subject control shift

- a. Jim_i asked Mary [PRO_i to be allowed to get himself a new dog].
- b. The children_i persuaded the teacher [PRO_i to be allowed to leave early].
- c. The council_i petitioned the mayor [PRO_i to be allowed to lower property taxes].
- d. John_i begged Mary [PRO_i to be allowed to consult a doctor].

(4) The complement does not have to be passivized

- a. Grandpa promised the children_i [PRO_i to be able to stay up for the late show].
- b. Montana_i was promised (by the doctor) [PRO_i to be healthy by game time on Saturday].
- c. Jim_i asked Mary [PRO_i to get himself a new dog].
- d. The children_i asked/persuaded/begged the teacher [PRO_i to leave early].

Intuitive generalization In the complement, semantically, the subject is not solely responsible for the situation (Farkas 1988), but rather the subject is permitted/prompted by another individual to be in a certain situation

2 The *de se* reading of control in centred-world semantics

2.1 Centered-world based analysis of *de se* attitude reports

- (5) a. Pavarotti wants to be famous. (*de se* report)

- b. Pavarotti wants that he will be famous. (*de se* not obligatory)
(Morgan 1970, Chierchia 1989)

(6) **Hintikka (1962)-style analysis:**

Attitude verbs quantify over possible worlds

$\llbracket (5a) \rrbracket^w = 1$ iff for all worlds w' compatible with Pavarotti's desires in w , Pavarotti is famous in w' .

(7) **Centered-world analysis** (*a la* Lewis 1979; Stephenson 2010 for a recent discussion):

Attitude verbs quantify over world-individual pairs (i.e. CENTERED WORLDS)

$\llbracket (5a) \rrbracket^w = 1$ iff for all those world-center pairs $\langle w', x' \rangle$ compatible with Pavarotti's desires in w , x' is famous in w' .

where CENTER-WORLD PAIRS COMPATIBLE WITH x 'S DESIRES IN w is the following set of individual-world pairs: $\{\langle w', x' \rangle: x$'s desires in w are met in w' and x identifies herself to be x' in $w'\}$

2.2 Compositional semantics

Semantic types for centered worlds

- (8) For $x \in D_e$ and $w \in D_w$, $\langle w, x \rangle \in D_c$

A control complement takes as its 1st argument a centered world

Putting aside the event argument, this is (the characteristic function of) a set of centered worlds. (The event argument will be needed later)

- (9) $\llbracket \text{PRO}_1 [t_1 \text{ to leave}] \rrbracket_e = \lambda w \lambda x \lambda e. \text{leave}(x, e, w)^1$
 $\uparrow \quad \quad \quad \rightsquigarrow \lambda \langle w, x \rangle \lambda e. \text{leave}(x, e, w)$ (un-Schönfinkelization)

Denotations of 2-place control predicates

- (10) $\llbracket \text{want} \rrbracket^w = \lambda P_{\langle c, vt \rangle} \lambda x. \forall \langle w', x' \rangle \in \text{WANT}_{w,x} \exists e [P(\langle w', x' \rangle, e)] : \langle \langle c, vt \rangle, et \rangle$

where $\text{WANT}_{w,x} := \{\langle w', x' \rangle: \text{the desires of } x \text{ in } w \text{ are met in } w' \text{ and } x \text{ identifies herself with } x' \text{ in } w'\}$ i.e. the set of centered worlds compatible with x 's desire in w

¹Here, PRO is assumed to be an element that obligatorily moves to the left edge of the complement and leaves a trace to be abstracted over, but has no semantic interpretation of its own. See Chierchia (1989) and Percus and Sauerland (2003) among others for similar views on PRO.

- (11) $\llbracket \text{Pavarotti wants PRO to leave} \rrbracket^w = 1$ iff for all centered worlds $\langle w', x' \rangle$ compatible with Pavarotti's desires in w , there is an event at w' such that x' leaves

How about 3-place predicate cases? (The denotations are to be revised!)

- (12) John promised the employees to leave. (subject control; *de me*)

- (13) John told/asked Mary to leave. (object control; *de te*)

- (14) $\llbracket \text{promise} \rrbracket^w = \lambda P_{\langle c, vt \rangle} \lambda y \lambda x. \forall \langle w', x', y' \rangle \in \text{PROM}_{w,x,y} \exists e [P(\langle w', x' \rangle, e)]$

$\text{PROM}_{w,x,y} := \{\langle w', x', y' \rangle: x$'s promises to y in w are met in w' and x identifies herself with x' and her addressee with y' in $w'\}$ i.e. the set of double-centered worlds compatible with what x promised to y in w

- (15) $\llbracket \text{ask} \rrbracket^w = \lambda P_{\langle c, vt \rangle} \lambda y \lambda x. \forall \langle x', y', w' \rangle \in \text{ASK}_{w,x,y} \exists e [P(\langle w', y' \rangle, e)]$

$\text{ASK}_{w,x,y} := \{\langle w', x', y' \rangle: \text{what } x \text{ asked to } y \text{ in } w \text{ is true in } w' \text{ and } x \text{ identifies herself with } x' \text{ and her addressee with } y' \text{ in } w'\}$ i.e. the set of double-centered worlds compatible with what x asked to y in w

- (16) $\llbracket \text{John asked Mary to leave} \rrbracket^w = 1$ iff for all double centered worlds $\langle w', x', y' \rangle$ compatible with what John asked to Mary in w , there is an event at w' such that y' leaves

3 Decomposing the denotations of 3-place control verbs

We decompose the denotations of 3-place control verbs into...

- (i) The general (lexically independent) component inducing a *de se* reading (regardless of whether it is *de me* or *de te*) for predicates of this semantic type
- (ii) The lexical variation among 3-place control verbs that determines whether the subject or the (indirect) object corresponds to the embedded agent.

3.1 The general meaning schema for 3-place control verbs

- (17) $\llbracket \text{R} \rrbracket^w = \lambda P \lambda y \lambda x. \exists \mathcal{F} \forall \langle w', x', y' \rangle \in \text{R}^*_{w,x,y} \exists e [P(\mathcal{F}(\langle w', x', y' \rangle), e)]$
where

- $\text{R}^*_{w,x,y} := \{\langle w', x', y' \rangle: \text{the content of the message/direction associated with } x$'s attitude $\llbracket \text{R} \rrbracket$ to y is true/met in w' and x identifies herself with x' and her addressee as $y'\}$

- $\mathcal{F} \in \{\lambda\langle w, x, y \rangle. \langle w, x \rangle, \lambda\langle w, x, y \rangle. \langle w, y \rangle, \lambda\langle w, x, y \rangle. \langle w, x \oplus y \rangle\}$
- The restriction on $\mathcal{F}$ captures the *de se* property of control i.e. the individual argument of the complement must be one of the matrix attitude centers
 - $\lambda\langle w, x, y \rangle. \langle w, x \rangle$: *de me* function
the attitude holder center is the individual argument of the complement
 - $\lambda\langle w, x, y \rangle. \langle w, y \rangle$: *de te* function
the addressee center is the individual argument of the complement
 - $\lambda\langle w, x, y \rangle. \langle w, x \oplus y \rangle$: *de nos* function
the attitude holder + addressee center is the individual argument of the complement

Support for the need of underspecified *de se* requirement

- (18) NEARLY-FREE CONTROL (Jackendoff and Culicover 2003)
- John_i talked to Sarah_j about PRO_{i/j/i+j/gen} taking better care of himself_i/herself_j/themselves_{i+j}/oneself_{gen}.
 - *Amy_k knows that John_i talked to Bill_j about PRO_k taking care of herself_k.
(long distance control)
 - *Brandeis_i is in a lot of trouble. John talked to Sarah about PRO_i firing the football coach.
(discourse antecedent control)
 - *John talked to Sarah about PRO_{i/j} undressing myself_i/yourself_j in public.
(speaker/hearer control)
- subject/object/subject+object split/generic control but no long distance/discourse control
 - Nearly-free control verbs...
 - obligatorily induce *de se* reading i.e. subject to the meaning postulate in (17)
 - but are underspecified as for the *de me/de te* lexical specification i.e. the choice of $\mathcal{F}$ is underspecified
 - The generic reading has a different structure from other readings, involving a DP complement, where the argument corresponding to PRO is already saturated.

3.2 Lexical variation among 3-place predicates

Agentivity Entailment (Lasnik and Fiengo 1974, Farkas 1988 and others)

- 3-place control verbs (except nearly-free control verbs) lexically entail that the controller argument must correspond to the *agent* of the event described by the complement²

(19) ?? John {promised/asked} Mary PRO to {be tall/resemble Sue/get hungry}.

The general schema in (17) + the agentivity entailment

- ‘Controller’ is an argument such that the lexical semantics of the control verb requires it to be the agent of the complement
- A ‘subject control’ verb is a control verb such that it requires its subject to correspond to the embedded agent. Thus, the lexical entry of *promise* is:

(20) $\llbracket \text{promise} \rrbracket^w = \lambda y \lambda P \lambda x. \exists \mathcal{F} \forall \langle w', x', y' \rangle \in \text{PROM}_{w,x,y} \exists e [P(\mathcal{F}(\langle w', x', y' \rangle), e) \wedge \text{Agt}(x', e)]$

- An ‘object control’ verb is a control verb such that it requires its object to correspond to the embedded agent. Thus, the lexical entry of *ask* is:

(21) $\llbracket \text{ask} \rrbracket^w = \lambda y \lambda P \lambda x. \exists \mathcal{F} \forall \langle w', x', y' \rangle \in \text{ASK}_{w,x,y} \exists e [P(\mathcal{F}(\langle w', x', y' \rangle), e) \wedge \text{Agt}(y', e)]$

Deriving the controller choice ($\mathcal{F}$) among its restriction from the Agentivity Entailment

- When the syntactic subject position of the complement (i.e. PRO) is occupied by the (embedded) agent
- (22) $\llbracket \text{PRO to leave} \rrbracket_e = \lambda \langle w, x \rangle \lambda e. \text{leave}(x, e, w) \wedge \text{Agt}(x, e)$
- since the embedded agent corresponds to the matrix controller argument (Agentivity Entailment)
 - the subject position of the complement (i.e. PRO) has the interpretation corresponding to the matrix controller argument
 $\Rightarrow$ **ordinary control**

Very roughly,...

- $\llbracket \text{PRO} \rrbracket =$ the embedded agent³
- the matrix controller argument $\simeq$ the embedded agent
- Therefore, $\llbracket \text{PRO} \rrbracket \simeq$ the matrix controller argument

²This does not hold for 2-place control verbs as in *John wants to be tall*.

³The bracket notation $\llbracket \text{PRO} \rrbracket$ here is not used in a technical way, but for the ease of presentation. In the actual compositional semantics I am proposing, PRO itself has no semantic interpretation.

How $\mathcal{F}$ is determined in actual examples

- (23) a. John_i promised Mary PRO_i to leave.
 b. John asked Mary_i PRO_i to leave.
- With a subject control verb, the individual argument of the complement corresponds to the matrix (underlying) subject.
 $\Rightarrow$ the value of $\mathcal{F}$ is $\lambda\langle w, x, y \rangle. \langle w, x \rangle$.
 - With an object control verb, the individual argument of the complement corresponds to the matrix (underlying) object.
 $\Rightarrow$ the value of $\mathcal{F}$ is $\lambda\langle w, x, y \rangle. \langle w, y \rangle$.

The denotations in (20), (28) + the derived choice of $\mathcal{F}$ wrt the complement

- (24) a. $\llbracket \text{promise} \rrbracket^w = \lambda y \lambda P \lambda x. \forall \langle w', x', y' \rangle \in \text{PROM}_{w,x,y} \exists e [P(\langle w', x' \rangle, e) \wedge \text{Agt}(x', e)]$
 b. $\llbracket \text{ask} \rrbracket^w = \lambda y \lambda P \lambda x. \forall \langle w', x', y' \rangle \in \text{ASK}_{w,x,y} \exists e [P(\langle w', y' \rangle, e) \wedge \text{Agt}(y', e)]$
- (25) $\llbracket \text{John asked Mary to leave} \rrbracket^w = \forall \langle w', x', y' \rangle \in \text{ASK}_{w,j,m} \exists e [\text{leave}(y', e, w') \wedge \text{Agt}(y', e)]$

But, what if the subject position of the complement does not correspond to the embedded agent?

4 Analyzing controller shift

4.1 When the embedded agent is not in PRO

- (26) John asked Mary PRO to be allowed to leave.
- (27) a. $\llbracket \text{allowed} \rrbracket^w = \lambda P \lambda x \lambda e. \exists y [\text{allowed}(x, P(x), e, w) \wedge \text{Agt}(y, e)]^4$
 b. $\llbracket \text{leave} \rrbracket = \lambda x. \text{leave}(x)$
 c. $\llbracket \text{PRO} \text{ } 1 \text{ } [\text{ } t_i \text{ } [\text{be} [\text{allowed} [\text{to leave}]]]] \rrbracket_e$
 $= \lambda w \lambda x \lambda e. \exists y [\text{allowed}(x, \text{leave}(x), e, w) \wedge \text{Agt}(y, e)]$
 $\leadsto \lambda \langle w, x \rangle \lambda e. \exists y [\text{allowed}(x, \text{leave}(x), e, w) \wedge \text{Agt}(y, e)]$

⁴I simplify the semantics of the embedded control verb *allow* here for expository purposes, but in theory, it too has the semantics involving quantification over centered worlds. My proposal does not hinge on this simplification.

Deriving $\mathcal{F}$ among its restriction via Agentivity Entailment

- The syntactic subject position of the complement (i.e. PRO) is not occupied by the embedded agent.
- Since, the matrix controller argument maps to the embedded agent (Agentivity Entailment).
- The subject position of the complement (i.e. PRO) cannot have the interpretation corresponding to the matrix controller argument.

Very roughly,...

- $\llbracket \text{PRO} \rrbracket \neq$ the embedded agent
- The matrix controller argument $\simeq$ the embedded agent
- Therefore, $\llbracket \text{PRO} \rrbracket \neq$ the matrix controller argument

The mapping of arguments has to be switched from the ordinary case

- $\mathcal{F}$ is chosen so that the controller argument does *not* map to the individual argument of the complement (i.e. the PRO argument).
- Due to the constraint on $\mathcal{F}$, this has to be the mapping that maps the non-controller matrix argument to the individual argument of the complement.
- In the case of *ask*, $\mathcal{F}$ is chosen so that the matrix object does *not* map to the individual argument of the complement: $\lambda\langle w, x, y \rangle. \langle w, x \rangle$

$$(28) \quad \llbracket \text{ask} \rrbracket^w = \lambda y \lambda P \lambda x. \exists \mathcal{F} \forall \langle w', x', y' \rangle \in \text{ASK}_{w,x,y} \exists e [P(\mathcal{F}(\langle w', x', y' \rangle), e) \wedge \text{Agt}(y', e)]$$

Example: *John asked Mary PRO to be allowed to leave*

- (29) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, x \rangle$: ‘subject control’ *de me*
 $\llbracket \text{John asked Mary PRO to be allowed to leave} \rrbracket^w = 1$ iff
 $\forall \langle w', x', y' \rangle \in \text{ASK}_{w,j,m} \exists e [\exists z [\text{allowed}(x', \text{leave}(x'), e, w') \wedge \text{Agt}(z, e)] \wedge \text{Agt}(y', e)]$
 $\Rightarrow$ **consistent specification of the embedded agent; Control shift!**
- (30) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, y \rangle$: ‘object control’ *de te*
 $\llbracket \text{John asked Mary PRO to be allowed to leave} \rrbracket^w = 1$ iff
 $\forall \langle w', x', y' \rangle \in \text{ASK}_{w,j,m} \exists e [\exists z [\text{allowed}(y', \text{leave}(y'), e, w') \wedge \text{Agt}(z, e)] \wedge \text{Agt}(y', e)]$
 $\Rightarrow$ **Inconsistent specification of the embedded agent**

- (31) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, x \oplus y \rangle$: ‘subject + object control’ *de nos*
 $\llbracket \text{John asked Mary PRO to be allowed to leave} \rrbracket^w = 1$ iff
 $\forall \langle w', x', y' \rangle \in \text{ASK}_{w,j,m} \exists e [\exists z [\text{allowed}(x' \oplus y', \text{leave}(x' \oplus y'), e, w') \wedge \text{Agt}(z, e)] \wedge \text{Agt}(y', e)]$
 $\Rightarrow$ **Inconsistent specification of the embedded agent**

4.2 Implicit ‘initiation’ in the complement

Control shift in cases where the complement is not passivized

- (32) a. Grandpa promised the children [PRO to be able to stay up for the late show].
 b. Montana_i was promised (by the doctor) [PRO_i to be healthy by game time on Saturday].
 c. Jim_i asked Mary [PRO_i to get himself a new dog].
 d. The children_i asked/persuaded/begged the teacher [PRO_i to leave early].

Implicit ‘initiation’ meaning introduced by the head INIT

The term ‘initiation’/‘initiator’ is from Farkas (1988)

- (33) $\llbracket \text{INIT} \rrbracket^w = \lambda P_{\langle s, vt \rangle} \lambda e. \exists e' [\text{initiate}(e, e', w) \wedge P(w)(e')]$
 (34) a. **initiate**(e, e', w) = 1 iff e' would not have happened without (i.e. is causally dependent on) e in w
 b. $\llbracket \text{INIT} \rrbracket^w(P)(e)$ is defined only when (in w) e has ‘control over’ any event that satisfies P . That is, any event that satisfies P would not have happened in w without (i.e. is causally dependent on) e .
 (35) $\llbracket \text{PRO} [1 [\text{INIT} [t_1 \text{ be able to stay up}]]] \rrbracket_e$
 $\uparrow$
 $= \lambda w \lambda x \lambda e. \exists e' [\text{initiate}(e, e', w) \wedge \text{able-to-stay-up}(x, e', w)]$
 $\leadsto \lambda \langle w, x \rangle \lambda e. \exists e' [\text{initiate}(e, e', w) \wedge \text{able-to-stay-up}(x, e', w)]$

Any choice of $\mathcal{F}$ is possible

- (36) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, x \rangle$
 $\llbracket (32a) \rrbracket^w = 1$ iff $\forall \langle w', x', y' \rangle \in \text{PROM}_{w,g,c} \exists e [\exists e' [\text{initiate}(e, e', w') \wedge \text{able-to-stay-up}(x', e', w')] \wedge \text{Agt}(x', e)]$
 ‘Grandpa promised the children to initiate the situation where he is able to stay up.’

- (37) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, y \rangle$
 $\llbracket (32a) \rrbracket^w = 1$ iff $\forall \langle w', x', y' \rangle \in \text{PROM}_{w,g,c} \exists e [\exists e' [\text{initiate}(e, e', w') \wedge \text{able-to-stay-up}(y', e', w')] \wedge \text{Agt}(x', e)]$
 ‘Grandpa promised the children to initiate the situation where the children are able to stay up.’ $\Rightarrow$ Control Shift
 (38) If $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, x \oplus y \rangle$
 $\llbracket (32a) \rrbracket^w = 1$ iff $\forall \langle w', x', y' \rangle \in \text{PROM}_{w,g,c} \exists e [\exists e' [\text{initiate}(e, e', w') \wedge \text{able-to-stay-up}(x' \oplus y', e', w')] \wedge \text{Agt}(x', e)]$
 ‘Grandpa promised the children to initiate the situation where both the children and he are able to stay up.’

- In other words, these cases end up the same as the Nearly-free control case.

Controller non-shift reading of ‘be allowed to’ examples

- (39) John_i asked Mary_j PRO_j to be allowed to leave.
 ‘John asked Mary to arrange things so that she would be allowed by some authority to leave.’
 (40) $\llbracket \text{PRO} [1 [\text{INIT} [t_1 \text{ be allowed to leave}]]] \rrbracket_e$
 $\uparrow$
 $= \lambda w \lambda x \lambda e. \exists e' [\text{initiate}(e, e', w) \wedge \text{allowed-leave}(x, e', w)]$
 $\leadsto \lambda \langle w, x \rangle \lambda e. \exists e' [\text{initiate}(e, e', w) \wedge \text{allowed-leave}(x, e', w)]$
 (41) $\llbracket (39) \rrbracket^w = 1$ iff
 $\forall \langle w', x', y' \rangle \in \text{ASK}_{w,j,m} \exists e [\exists e' [\text{initiate}(e, e', w') \wedge \text{allowed-leave}(y', e', w)] \wedge \text{Agt}(y', e)]$
 (Using $\mathcal{F} = \lambda\langle w, x, y \rangle. \langle w, y \rangle$)

4.3 Summary

- When the complement’s syntactic subject position (i.e. PRO) corresponds to the embedded agent, ordinary control pattern arises through the Agentivity Entailment.
- When the complement’s syntactic subject position (i.e. PRO) does *not* correspond to the embedded agent, the only possible mapping by $\mathcal{F}$ is to map the non-agent individual argument (i.e. in PRO) of the complement to the non-controller argument. $\Rightarrow$ Controller Shift

5 Predictions

Correlation between the agentivity of PRO and controller shift

- Controller shift is possible only when the embedded agent is associated with a syntactic position that is different from PRO.
- This naturally explains why CS is possible with passivized complements.
- Also, given INIT, this explains the intuition that CS occurs when the subject of the complement is not solely responsible for the situation (Farkas 1988), but the situation is somehow ‘initiated’ by another individual

The role of (ordinary) controller argument in the complement

- In controller shift, the ordinary controller must correspond to the agent of a complement event even if it is implicit.
- *John asked Mary to be allowed to leave* never has a controller-shifted reading where Mary is not the initiator/allower.

Obligatoriness of *de se* reading in controller shift

- CS sentences must require a *de se* reading.
- Prediction: (42a) (in contrast to (42b)) must have a *de te* attitude toward the allowee and *de me* attitude toward the (implicit) initiator/allower.

- (42) a. John promised Mary PRO to be allowed to leave.
 b. John promised Mary that she will be allowed to leave.

Situation 1 (*De me* but not *de te*) John is a leader of a project team in a company. One day, John heard from someone that an employee called Mary was hoping to leave the team. John said to a random employee “Mary will be allowed to leave”, without knowing that the person is actually Mary.
 ⇒ (42a) False; (42b) True

Situation 2 (*De te* but not *de me*) John, the project leader, became amnesiac, and mistakenly thought that he was a fortune teller. Hearing Mary’s complaint, John, as a fortune teller, said to Mary: “Your leader will allow you to leave.”
 ⇒ (42a) False; (42b) True

6 Conclusions

- Distinguishing between (i) the *de se* meaning and (ii) the lexical entailment about the embedded agent in the denotations of control verbs provides a

correct account of controller choice including control shift.

- The current analysis gives a new piece of evidence for the centered-world analysis of attitude ascription (at least in the case of control) from the point of view of controller choice.

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